

Meghan E. Walsh

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PROFESSIONAL EXPERIENCE

Clinical Research Coordinator

May 2023 – Present

Boston Children's Hospital, Neurology Department, Boston MA

- Developed and implemented a real-time neurofeedback pipeline:
 - Designed a real-time fMRI neurofeedback pipeline using Python, Java, and shell scripting to compute and display in-scanner brain activation for participants with ADHD. This project aimed to upregulate attentionally-implicated brain regions and test the feasibility of fMRI neurofeedback as a drug-free treatment for ADHD.
 - Developed a video game interface that translated real-time activation levels into controllable in-game actions to support self-neuromodulation training.
 - Built a portable Docker environment encapsulating all dependencies and custom software.
- Contributed to the development of an intra-volume motion characterization tool for fMRI data:
 - Developed a motion-characterization tool which detects intra-volume motion using slice-level registration strategies in clinical datasets (ADHD, ASD, TD cohorts) via a Python and shell script-based, docker-containerized pipeline.
 - Implemented performance improvements in Python scripts using multithreading and multiprocessing, reducing runtime by over half.
 - Implemented optimization strategies (Simplex, ADAM) and generated mutual information landscapes across rotation/translation sweeps to evaluate fMRI image registration accuracy, producing multidimensional MI plots to identify optimal transform parameters.
 - Produced figures for grant applications that demonstrate intra-volume motion spikes, displacement traces, and effects of motion scrubbing and reconstruction on GLM Z-statistics and R^2 maps.
- Conducted neuroimaging data analyses:
 - Performed preprocessing and functional connectivity analysis using ABCD normative connectomes, conducted permutation-based statistical tests, and analyzed meta-analysis data through lesion network mapping, overlap metrics, and VLSM analyses with PALM to examine the effects of stimulant use on brain structure and activation in individuals with ADHD.
- Coordinated IRB protocols and managed the ethical handling of research data.

Undergraduate Research Assistant

June 2021 - May 2023

Boston University, Boston MA

- Participated in designing and piloting psychophysical experiments investigating higher-order visual awareness, attention, and perception and collected behavioral and eye-tracking data from participants.
- Gained hands-on experience with MATLAB and SPSS for stimulus presentation, data acquisition, and data analysis of reaction time and accuracy data, as well as operating eye-tracking technology to examine attentional and perceptual processes.
- Delivered presentations on research findings and methodology.
- Developed foundational skills in data analysis and utilization of technology to enhance the understanding of cognitive processes, including attention and visual awareness.

Substitute Teacher

May 2021 - May 2023

Lincoln Public Schools, Lincoln MA

- Substituted in preschool through middle school classrooms 3–5 days a week, managing groups of up to 40 students. Worked one-on-one with students with special needs, including ADHD, autism, and learning disabilities, following individualized plans and providing tailored academic and behavioral support.

Preschool Assistant Teacher

July 2021 - August 2021

Lincoln Public Schools, Lincoln MA

- Supported special-needs students in an extended school-year program, aiding those with learning disabilities, autism, ADHD, and physical or language barriers
- Guided preschoolers in writing and reading, facilitated emotional regulation activities, and prepared instructional material.

EDUCATION

Bachelor of Arts in Psychology

Graduated May 2023

Boston University

- Relevant Coursework
 - Introduction to Databases and Data Mining
 - Experimental Design (Statistics for Psychology Research)
 - Cognitive Neuroscience
- Awards and Honors
 - Dean's List (Every Semester, 2015–2019)
 - Graduated magna cum laude
 - Psi Chi International Honor Society in Psychology
- Extracurricular Experience
 - Treasurer of Active Minds Club (September 2021 to May 2023)
- GPA: 3.83, Major GPA: 3.97

PROFESSIONAL DEVELOPMENT & CONTINUING EDUCATION

Southern New Hampshire University (SNHU)

March 2025 - April 2025

Individual Coursework Student

- Operating Platforms (CS230)
- Discrete Mathematics (MAT230)

IBM Online Courses Via Coursera

June 2024 - July 2024

Completed Online Courses

- Introduction to DevOps
- Introduction to Cloud Computing
- Introduction to Agile Development and Scrum
- Hands-on Introduction to Linux Commands and Shell Scripting
- Getting Started with Git and GitHub

GRANTS AND AWARDS

InterfaceNeuro Conference Travel Award - Georgia Institute of Technology, Atlanta, GA, USA May 2025
Development and Implementation of a Portable Real-Time fMRI Neurofeedback Protocol

Received a \$500 travel award to present my real-time fMRI neurofeedback pipeline at InterfaceNeuro, an interdisciplinary neurotechnology conference focused on advancing brain-body-machine interfaces.

Steve Samuels & Ami Cipolla Innovation Pilot Grant - Boston Children's Hospital, Boston, MA, USA April 2024
Optimizing Real-time fMRI Neurofeedback to Improve Sustained Attention in ADHD

Received a one-year, \$100,000 pilot grant funding innovative research projects exploring neurobiological mechanisms of attention and anxiety and offering translational potential to develop new biomarkers or therapeutic strategies for conditions like ADHD.

Undergraduate Research Opportunity Program Grant - Boston University, Boston, MA, USA May 2022
Dissociating subjective and objective awareness reports using priming

Awarded a stipend through Boston University's Undergraduate Research Opportunity Program (UROP) to conduct faculty-mentored research during the summer term gaining hands-on experience in study design, data analysis, and academic presentation.

PUBLICATIONS

Tian, K. J., Maniscalco, B., Epstein, M. L., Shen, A., Castaneda, O. G., Kurosawa, T., Motzer, J. A., Olsson, E., Russell, E. E., **Walsh, M. E.**, Wang, J., Awrang Zeb, T. B., Brown, R., Lamme, V. A. F., Lau, H., He, B. J., Brascamp, J. W., Block, N., Chalmers, D., ... Denison, R. N. (2025). When awareness outstrips performance: critical tests of subjective inflation under inattention. *bioRxiv*, 2025.07.03.661972. <https://doi.org/10.1101/2025.07.03.661972> (preprint)

Vision Sciences Society Annual Meeting Published Abstract September 2024

Tian, K., Maniscalco, B., Epstein, M., Shen, A., Graham Castaneda, O., Arzu, G., Kurosawa, T., Motzer, J., Olsson, E., Romero, L., Russell, E., **Walsh, M.**, Wang, J., Bek Awrang Zeb, T., Brown, R., Lamme, V., Lau, H., He, B., Brascamp, J., ... Denison, R. (2024). Attention robustly dissociates objective performance and subjective visibility reports. *Journal of Vision*, 24(10), 408–408. <https://doi.org/10.1167/jov.24.10.408>

- Talk: 27th Annual Meeting of the Association for the Scientific Study of Consciousness (ASSC), Tokyo, Japan, July 5, 2024. Presented by K. Tian.
- Talk: Vision Sciences Society (VSS) Meeting, May 18, 2024. Presented by K. Tian.
- Talk: 26th Annual Meeting of the Association of Scientific Studies of Consciousness (ASSC), New York, NY (NYU), June 23, 2023. Presented by K. Tian.

Vision Sciences Society Annual Meeting Published Abstract August 2023

Tian, K., **Walsh, M.**, & Denison, R. (2023). Dissociating subjective and objective awareness reports using priming. *Journal of Vision*, 23(9), 4737–4737. <https://doi.org/10.1167/jov.23.9.4737>

- Poster: Vision Sciences Society (VSS) Meeting, St. Pete, FL, May 21, 2023. Presented by K. Tian.
- Poster: Boston University Undergraduate Poster Symposium, Boston, MA, October 21, 2022. Presented by M. Walsh.

POSTERS

InterfaceNeuro Conference Poster - Georgia Institute of Technology, Atlanta, GA, USA

May 2025

Development and Implementation of a Portable Real-Time fMRI Neurofeedback Protocol

Presented a poster at InterfaceNeuro showcasing a portable real-time fMRI neurofeedback protocol integrating rapid neuroimaging analysis and deployable software tools for in-scanner neural feedback.

Undergraduate Research Opportunity Program Poster - Boston University, Boston, MA, USA

October 2022

Dissociating subjective and objective awareness reports using priming

Discussed findings from UROP-funded research, highlighting the differentiation of subjective perception and objective performance measures of visual awareness.